Julia Ebert

Robotics Research & Development • Software Engineering • Boston, MA

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Skills

Computer Science Algorithm development • C++ • Python • Linux • Docker • Git/version control • ZeroMQ • Protocol buffers (Protobuf) • Robot Operating System (ROS) • MuJoCo • JavaScript • HTML/CSS

Engineering Computer-aided design (OnShape) • 3D printing • Basic PCB design (Eagle) • Laser cutting

Experience

Boston, MA 2023 –
Fleet Robotics
Autonomy Lead

- Architect and implement the software system from the ground up, including hardware selection, communication protocols (ZMQ, protobuf), DevOps (Docker, CI/CD pipeline), and user interface development, enabling Fleet's first on-ship robot demonstrations.
- Manage a team of three software engineers and supervise high school through PhD student interns.
- Design and implement robot control systems, safety mechanisms, and a robust logging system.
- Develop robust path planning and execution systems for autonomous underwater navigation on 3D curved steel surfaces, addressing challenges of limited sensing and harsh operating conditions.
- Lead software project planning, collaborating with product management and executives to align goals, manage timelines, and mitigate risks.

Brighton, CO 2022 – 2023
Outrider
Software Engineer, Mission Planning (Remote)

- Spearheaded design and development (C++, ROS) of new multi-robot planning for Outrider's of autonomous distribution yard trucks.
- Led cross-functional project teams to create new robot behaviors toward product goals.
- Supported test site and customer deployments of the mission planning system.

Cambridge, MA 2016 – 2022
Harvard University Self-Organizing Systems Research Group, Prof. Radhika Nagpal
Doctoral Researcher

- Developed collective spatial decision-making algorithms for simulated and physical robot collectives. Includes bio-inspired and Bayesian decision and movement algorithms.
- Created open-source Kilosim (C++) to simulate hundreds of robots at up to 1000x real time.
- Collaborated with MIT Media Lab to create heterogeneous robot swarm for inspection on space stations, including algorithm development and hardware testing in microgravity (Zero-G flights).
- Supervised undergraduate and masters student research projects, and taught robotics and digital fabrication courses.

Livermore, CA Summer 2018
Lawrence Livermore National Laboratory, Dr. Michael Schneider
Computational Science Research Intern

- Programmed, refactored, and documented research codebase (Python) for space situational awareness (SSA), since used extensively by researchers at LLNL and in review for JOSS publication.
- Developed a simulator and visualization tools (Python) for orbit observation by low earth orbit satellites.

Education

Cambridge, MA 2022
Harvard University
PhD in Computer Science

- Thesis: *Distributed Decision-making for Inspection by Autonomous Robot Collectives*
- Department of Energy Computation Science Graduate Fellow (DOE CSGF) • Siebel Scholar

London, UK 2016
Imperial College London
Master of Research (MRes) in Bioengineering, with Distinction

- Marshall Scholar • Thesis: *Assisting Balance Recovery with a Lower Limb Exoskeleton*

Boston, MA 2015
Northeastern University
BS in Behavioral Neuroscience, Minor in Computer Science

Select Publications

J Ebert et al. (2022) A Hybrid PSO Algorithm for Multi-robot Target Search and Decision Awareness. *IROS 2022.*

J Ebert et al. (2020) Bayes Bots: Collective Bayesian Decision-Making in Decentralized Robot Swarms. *ICRA 2020.*

J Ebert et al. (2018) Multi-feature collective decision making in robot swarms. *AAMAS 2018.*